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MILITARY SPECIFICATION

AIR CONDITIONER, TRAILER-MOUNTED, FREON CYCLE, 11-1/2 TON REFRIGERATING CAPACITY, TYPE MA-3

1. SCOPE

1.1 This specification covers one type of freon-cycle, trailer-mounted air conditioner having a nominal cooling capacity of 138,000 BTU/hr (11-1/2 tons), designated Type MA-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents form a part of this specification. The applicable issues of the Governmental documents shall be the issues in effect on the date of invitation for bids. The applicable issues of the nongovernmental documents shall be those available in the files of the procuring activity and which are made a part of the invitation for bids.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-561	Solder, Silver
TT-E-489	Enamel, Alkyd, Gloss, Synthetic (For Exterior And Interior Surfaces)
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting Lacquer-Resisting
UU-P-271	Paper, Wrapping, Waterproof Kraft
VV-L-825	Lubricating Oil, Refrigerant Compressor
WW-T-799	Tubing, Copper, Seamless (For Use With Solder-Joint or Flared-Tube Fittings)
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof -- For Packaging and Sealing

Military

JAN-P-100	Packaging and Packing for Overseas Shipment, General Specification
MIL-C-104	Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of

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MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-H-4607	Heater, Engine and Shelter, Ground, Type H-1
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For
MIL-E-5086	Wire, Electrical, 600-Volt, Copper, Aircraft
MIL-C-5541	Chemical Films For Aluminum And Aluminum Alloys
MIL-L-7178	Lacquer, Cellulose Nitrate, Gloss, For Aircraft Use
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification For
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-T-9107	Test Reports, Preparation of
MIL-L-21260	Lubricating Oil, Internal-Combustion Engine, Preservation
MIL-E-26248	Engines, Gasoline, Ground Use, Packette, General Specification For
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-D-70327	Drawings, Engineering And Associated Lists

STANDARDS

Federal

FED-STD-182	Continuous Identification Marking of Nickel And Nickel Base Alloys
FED-STD-183	Continuous Identification Marking of Iron And Steel Products
FED-STD-184	Identification Marking of Aluminum, Magnesium, And Titanium
FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-143	Specifications And Standards Order of Precedence for the Selection of
MIL-STD-185	Continuous Identification Marking of Copper And Copper Base Alloys
MS24531	Battery-Storage, Nickel-Cadmium 70-Amp Hr

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MS24559 Engine-Gasoline, Type GLU-3/U 110 HP Packette (Formerly PE 90)
MS33562 Connection, Aircraft Ground Air Conditioning, 8 Inch, Minimum Requirements
MS33586 Metals, Definition of Dissimilar
MS35000 Battery, Storage, Lead-Acid, Waterproof, For Tactical Vehicles

Air Force-Navy Aeronautical

AN2552 Receptacle - External Power, 28 Volt DC

DRAWINGS

Air Force

49B6564 Ring - Eyebolt, 25,000 Lb Cargo Tiedown

Recony Corporation

111,000 Index for Drawing of Air Conditioner, Trailer Mounted, Freon Cycle, 11-1/2 Ton Refrigerating Capacity, Type MA-3

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society Of Mechanical Engineers

Power Test Code

19.5:4 Supplements on Instruments and Apparatus; Part 5, Chapter 4, Flow Measurement by Means of Standardized Nozzles and Orifice Plates

(Copies of the ASME power test code may be obtained from the American Society of Mechanical Engineers, 29 West 39th Street, New York 18, New York.)

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American Standards Association Bulletins

Z24.3	Sound Level Meters
Z24.10	Octave Band Filter Set

(Copies of the ASA bulletins may be obtained upon application to the American Standards Association, 70 East 45th Street, New York, New York.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The air conditioner shall consist of the following major components and the necessary accessory equipment:

Description	See Requirement
a. Prime mover	3.8
b. Compressor	3.9
c. Condenser	3.10
d. Cooling-air fan	3.11
e. Evaporator	3.12
f. Conditioned-air blower	3.13
g. Refrigerant receiver	3.14
h. Refrigeration accessories	3.15
i. Cabinet	3.16
j. Ducts	3.17
k. Air filters	3.18
l. Fuel system	3.19
m. Trailer mobility provisions	3.20
n. Control system and instrumentation	3.21
o. Accessory equipment	3.22

3.3 General requirements

3.3.1 Recony drawing index.- The requirements of the drawings listed in Recony Index 111,000 apply as requirements of this specification. (See 6.2.) The requirements specified in this specification are definitive of the requirements covered by the details on the drawings. When the requirements of the Recony drawings and this specification conflict, this specification shall govern. The government-loaned sample (see 3.31.1) shall be used by the contractor as a guide for engineering purposes only in producing the air conditioner.

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3.3.2 General specification.- The requirements of MIL-S-8512 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.4 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as specified in 3.4.1 and 3.4.2.

3.4.1 Commercial parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS and AN) without alteration

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.4.2 Standard parts.- With the exceptions specified in 3.4.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.5 Materials

3.5.1 Protective treatment.- When materials are used in the construction of the air conditioner that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5.2 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to salt spray or atmospheric conditions likely to be met in storage or normal service.

3.5.2.1 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

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3.6 Design and construction.- The air conditioner shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.6.1 Maintenance.- The air conditioner shall be so constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases.- The air conditioner shall be so designed and constructed that maintenance can be accomplished with the minimum use of special tools.

3.6.2 Foolproofness.- Where practical, installation of systems, components, and parts shall be as simple and foolproof as possible to prevent maintenance personnel from inadvertently reversing or mismating fittings and couplings, fuel, oil, and refrigerant lines, airflow ducts, mechanical linkages, instrument leads, and electrical connections.

3.6.3 Life.- The air conditioner shall be designed for a minimum operating life of 1,000 hours between major overhauls and shall be capable of operating for 500 hours at design capacity without failure of any major part.

3.6.4 Refrigerant.- The air conditioner shall be designed for use of dichlorodifloromethane (Freon 12, or equal).

3.6.5 Hoisting and tiedown provisions.- Tiedown provisions shall be provided and shall serve as both tiedown and hoisting provisions. The hoisting and tiedown provisions shall include four tiedown ring assemblies conforming to Drawing 49B6564. These provisions shall be capable of withstanding a vertical acceleration of 2g. The tiedown ring assemblies shall be located around the periphery of the unit in a horizontal plane. The location of the ring assemblies shall be further governed by hoisting considerations. For hoisting, the rings shall be located with respect to the center-of-gravity in order to comply with the following requirements:

a. The air conditioner shall not tilt at an angle greater than 15° when hoisted from a single point with four equal-length cables attached to the ring assemblies.

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3.6.5.1 Single-point suspension over the center-of-gravity may be used as an alternate method of hoisting provided the 2-g requirement is met.

3.6.6 Acceleration forces.- All components of the air conditioner shall be constructed to withstand acceleration forces of 3g along each of the major axis in either direction and shall be capable of satisfactory operation as specified herein after such exposure. The air conditioner shall be of such construction that all components will be restrained from causing any external damage in the event that acceleration forces of 8g along the horizontal axes in either direction or 4-1/2g downward are encountered. Tiedown provisions shall not fail under these forces; however, the air conditionér need not be operational after such exposure. Ability of the dry air conditioner (see 3.6.6.1) to withstand these acceleration loadings shall be supported by engineering data.

3.6.6.1 Dry unit.- For the purpose of this specification dry unit or dry air conditioner will mean without fuel but not drainage of oil or refrigerant.

3.6.7 Noise level.- The noise level generated by the air conditioner shall not exceed the following values when measured as specified in 4.6.7:

Frequency Bands (CPS)	Octave Band Pressure Level (In Decibels Ref. 0.0002 Microbar)	
	Operator	Average at 25 feet
37- 75	106	91
75- 150	96	81
150- 300	88	73
300- 600	85	70
600-1,200	84	69
1,200-2,400	83	68
2,400-4,800	82	67
4,800-9,600	81	66

3.6.8 Radio interference.- The air conditioner shall be so shielded against radiated radio interference that the radiated interference will not exceed the limits specified in MIL-I-26600.

3.6.9 Detachable fittings.- Wherever practicable, disconnect plugs and receptacles in the electrical system and readily detachable- and attachable-type fittings in mechanical components shall be used to permit rapid component removal and replacement.

3.6.10 Locking devices.- All screws, pins, bolts, et cetera, shall be furnished with USAF standard locking devices. Safety wire, self-locking nuts, cotter pins, and lockwashers will be acceptable.

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3.6.11 Common parts.- Maximum practicable use shall be made of interchangeable hardware and fastening devices. The number of types and sizes of bolts, screws, nuts, washers, and similar common parts shall be held to a minimum.

3.6.12 Lubrication.- The components of the air conditioner shall be so lubricated that they will not require frequent servicing. The capacity of sumps or reservoirs, if used, shall be sufficient for 50 hours of operation. Seals shall be provided to prevent contamination of discharged conditioned air with oil from the air-conditioner lubrication system.

3.6.13 Wiring and cables.- All wiring and cables used in the electrical circuit shall conform to MIL-W-5086.

3.6.14 Refrigerant piping.- All refrigerant piping shall be type L copper tubing conforming to WW-T-799 except as specified in 3.6.14.1. The tubing shall be fabricated with the use of high-temperature solder conforming to QQ-S-561.

3.6.14.1 Flexible connections.- The refrigerant piping system shall be equipped with flexible connections to reduce vibration to a minimum consistent with good engineering practice. A minimum of two connections shall be provided, one at the compressor outlet and one at the compressor inlet.

3.6.15 Expansion valve.- A thermostatic-type expansion valve of adequate capacity shall be installed in the refrigerant supply connection to the evaporator. The expansion valve shall be provided with a liquid-refrigerant strainer that may be an integral part of the valve.

3.7 Performance

3.7.1 Ambient operating range.- The cooling system shall be capable of operating within the range of ambient conditions described on a psychrometric chart by the following limits:

a. The 0° F dry-bulb to intersect with its saturation line.

b. The saturation line from 0° to 90° F

c. A line from 90° F dewpoint on the saturation line to 140° F at 0 percent relative humidity.

3.7.1.1 In addition, the prime mover and component assemblies in operation when the air conditioner is used to deliver unrefrigerated

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air shall be capable of starting and operating satisfactorily throughout ambient temperatures ranging from 0° to -65° F.

3.7.2 Basic rating conditions and delivery.- At sea-level barometric pressure, the air conditioner shall be capable of delivering specified quantities of conditioned air under the conditions specified in 3.7.2.1, 3.7.2.2, 3.7.2.3, and 3.7.2.4. The basic refrigeration capacity of the air conditioner shall be 138,000 BTU/hr when operating at ambient temperatures of 100° F dry bulb (DB) and 85° F wet bulb (WB) and delivering air at 50° F at the end of the 25-foot duct specified herein. At the basic rating condition, the air conditioner shall deliver 75 pm of air against static pressures ranging from 0 to 3 psig.

3.7.2.1 Design load.- Under summer-day ambient conditions of 100° F DB and 76° F WB, the air conditioner shall be capable of delivering 100 pm of 45° F air against static pressures ranging from 0 to 3 psig.

3.7.2.2 Hot-day ambient conditions.- When operating in ambient conditions of 125° F DB and 75° F WB, the air conditioner shall be capable of delivering 80 pm of 45° F air against static pressures ranging from 0 to 3 psig.

3.7.2.3 Humid-day conditions.- When operating in ambient conditions of 97° F DB and 87° F WB, the air conditioner shall be capable of delivering 70 pm of 45° F air against static pressures ranging from 0 to 3 psig.

3.7.2.4 Cold-day ambient conditions.- When operating at an ambient dry-bulb temperature of 45° F , the air conditioner shall be capable of delivering 100 pm of air at a temperature between 35° and 45° F against static pressures ranging from 0 to 3 psig.

3.7.2.5 Airflow regulation.- The air conditioner shall be provided with a means of regulating airflow so that the air discharge flow rate can be manually controlled between 20 and 100 percent of that specified in 3.7.2.1 with the engine operating at maximum speed (2,550 rpm). The discharge air temperature shall be manually variable and automatically maintained (±3° F at full load, ±5° F at reduced load) at any temperature selected by the operator between 35° and 65° F DB at any condition of airflow rate or refrigeration load. However, at airflows less than 50 pm and ambient temperatures exceeding 100° F DB, the air conditioner shall deliver air at a temperature above coil frosting but not exceeding 50° F. Temperature setting shall be by means of a single selector located on the control panel. The direction of selector movement necessary to increase the temperature shall be indicated.

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3.7.3 Tilted position.- The air conditioner shall be capable of operating satisfactorily when its normal operating plane is at an angle of 8-1/2° in any direction from the true horizontal.

3.7.4 Maneuverability.- The trailer shall be capable of executing full right and left turns while being towed at speeds of from 8 to 10 mph. The trailer shall not capsize nor skid during turns on dry concrete or macadam surfaces. The trailer shall be capable of turning within a radius of 17-1/2 feet.

3.7.4.1 Obstructions.- The air conditioner shall be towable over 1-inch-high obstructions.

3.7.5 Environmental conditions.- The air conditioner shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Storage temperatures ranging from -65° to +160° F

b. Radiant energy as found under natural conditions

c. Exposure to salt-sea atmosphere

d. Fungus growth as encountered in tropical climates

e. Rainfall as encountered in any locale

f. Sand and dust particles as encountered in desert areas.

3.8 Prime mover.- The air conditioner shall be driven by a GLU-3/U Packette engine conforming to MIL-E-26248 and MS24559. The engine shall operate at a governed speed of 2,550 rpm.

3.8.1 Winterization provisions.- Winterization provisions shall not be included with the prime mover. However, the air conditioner design and prime mover installation shall be such that starting at ambient temperatures of -65° F can be accomplished with a Type H-1 heater conforming to MIL-H-4607 (see 3.31.1) after preheating for a maximum of 30 minutes.

3.8.2 Clutches.- Provisions shall be made for disengaging all driven components except the generator and conditioned air blower from the prime mover for ease of starting during low ambient temperatures and during periods of self-mobility.

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3.8.3 Direct-current generator.- All electrical devices and controls shall be operated on 24V, d-c electrical power to be provided by a direct-current generator mounted directly on and driven by the prime mover.

3.9 Compressor.- The compressor shall be of a type conforming to good refrigeration practice and shall have sufficient capacity to meet the air-conditioner capacity requirements specified herein when operating at allowable speeds for the prime mover. The compressor shall be directly driven by the prime mover.

3.9.1 Lubrication system.- The compressor shall be lubricated by means of an integral or attached lubrication system. The lubricant shall be of a type conforming to VV-L-825.

3.9.1.1 Oil reservoir.- An oil reservoir shall be included in the system of sufficient capacity for lubrication of the compressor for at least 500 hours.

3.9.1.2 Oil level gage.- An oil level gage shall be provided.

3.10 Condenser.- The condenser shall be of the air-cooled extended-surface type with copper fins and copper tubes. The fins shall be securely bonded to the tubes by a metallic bond such as is formed by hot-dipping, soldering, or brazing. The condenser shall be capable of withstanding an internal pressure of at least 500 psig without rupture or distortion. It shall have adequate surface to enable the unit to meet the maximum capacity requirement specified in 3.7.2.1 with a refrigerant condensing temperature not exceeding 130° F (head pressure of approximately 180 psig).

3.11 Cooling-air fan.- The air conditioner shall be equipped with a cooling-air fan or fans employed in such a manner as to draw ambient air through the condenser and reject the heated air through the top of the air conditioner in a vertical direction so that the heated discharge air will not recirculate through the air conditioner. In addition, air warmed by the condenser shall not pass over the prime mover or over other operating components of the air conditioner in such a manner that performance or efficiency of the air conditioner will be impaired. The cooling-air fan shall be driven by the prime mover. The use of a V-belt will be permissible.

3.12 Evaporator.- The evaporator shall be of the extended-surface type with copper fins and copper tubes. The fins shall be securely bonded to the tubes by a metallic bond such as is formed by hot-dipping, soldering, or brazing. The evaporator shall be capable of withstanding an internal pressure of at least 500 psig without

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rupture or distortion and shall be equipped for drainage of condensate water overboard. The evaporator shall have adequate surface, and tube distribution within the coil shall be so arranged that the suction pressure at the outlet of the coil will exceed 28 psig when the unit is operating at the maximum capacity requirement specified in 3.7.2.1.

3.13 Conditioned-air blower.- The conditioned-air blower shall be driven by the prime mover and shall be capable of discharging the conditioned air against a static pressure of 3 psig at the end of the 25-foot flexible duct section specified herein when operating under the conditions specified in 3.7.2. A V-belt drive of the blower from the prime mover will be permissible. If the blower employed creates a pressure greater than 4-1/2 psig when airflow is restricted or blocked off, a pressure relief valve shall be employed in the system to prevent overload of the prime mover or blower drive mechanism or both. The relief valve shall be set to start relieving air at 4 psig and shall pass the total airflow of the blower at a pressure not greater than 5 psig.

3.14 Refrigerant receiver.- The air conditioner shall be equipped with a refrigerant receiver of adequate capacity to hold the required charge of refrigerant for operation. The receiver shall be equipped with a safety relief valve or a fusible safety plug.

3.15 Refrigeration accessories

3.15.1 Charger.- A means shall be provided for charging the refrigeration system without the use of special equipment.

3.15.2 Dryer and strainer.- A refrigerant dryer and strainer shall be provided. The dryer and strainer shall have a replaceable drying agent.

3.15.3 Sight glass.- A sight glass shall be so provided in the liquid refrigerant line from the receiver that the condition of the refrigerant may be observed.

3.15.4 Purge valve.- A purge valve or port shall be provided for purging air from the refrigerant system. The valve shall be located at or near the highest point in the inlet header to the condenser.

3.16 Cabinet.- The air conditioner shall be fully enclosed in a lightweight metallic cabinet having sufficient strength and rigidity to withstand the mobility requirements specified herein. The cabinet shall be designed to present a neat, smooth external appearance free from sharp corners and edges. All major components and working parts of the air conditioner shall be readily accessible for maintenance and replacement through convenient doors or removable panels.

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3.16.1 Doors.- All cabinet exterior openings that may be open during air-conditioner operations shall be provided with weather-tight doors for weather sealing such openings when the air conditioner is not in operation. All doors shall be designed to present a near-flush exterior of the cabinet when the doors are in the open position. Doors hinged at the top, that may fall and cause injury to operating or maintenance personnel, shall be avoided. The number of doors shall be held to a minimum.

3.16.1.1 Latches.- All doors shall have latches for holding them securely in the closed position.

3.16.1.2 Fixtures.- Doors hinged at the side shall be equipped with fixtures that will hold the doors firmly open so that propeller and jet blasts or strong winds will not cause the doors to close and injure personnel.

3.16.1.3 Door handles.- Door handles shall be as flush as possible to provide a neat appearance. They shall be convenient for use by an operator wearing arctic gloves.

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3.16.2 Weather resistance.- The cabinet shall be constructed to resist weather and to prevent rain, water, sand, and dust from collecting within the cabinet in quantities that would be detrimental to proper operation of the air conditioner.

3.16.3 Discharge and intake openings.- The cabinet shall be provided with three openings on its external walls. One opening shall be the intake for the air to be conditioned and the other two shall be the discharge openings for the conditioned air. Each of the two discharge openings shall incorporate a male coupling with provisions for attachment of the ducts specified herein. Delivery of the total amount of conditioned air shall be possible through either outlet, or both.

3.16.3.1 Cover.- A captive cover shall be provided for blanking off one outlet for single-duct operation.

3.16.4 Duct storage space.- The cabinet shall contain a space for storing the ducts specified herein when they are not in use.

3.17 Ducts

3.17.1 Internal ducts.- If required, air ducts shall be provided in the cabinet for guiding air through the various assemblies. The ducts shall be constructed of lightweight material and all bends and turns shall be internally smooth. Where applicable, duct intersections shall insure proper uniform mixing of the various airstreams.

3.17.2 Flexible air ducts.- The air conditioner shall be equipped with three insulated air ducts each 25 feet in length. The ducts shall be of 8-inch nominal diameter and shall incorporate a male connection on one end that will connect to the female 8-inch aircraft air conditioning connection in accordance with MS33562. The other end of the duct shall incorporate a female connection in accordance with MS33562 so that the ducts can be connected in series. The female end of the ducts shall connect to the air conditioner discharge opening. The ducts shall withstand an internal air pressure of 6 psig and a tensile pull of 150 pounds when the force is applied to the end connections.

3.17.2.1 Seal.- The seal shall be a part of the male coupling and shall be of a material that will prevent leakage at the joint. If a fixed, claw-type male coupling is used, the seal shall be properly protected against blowout by a pressure up to 6 psig and shall be of a consistency that will permit easy connection to a fixed position mating coupling by one man. Male couplings employing movable claws actuated by cam or eccentric mechanisms and handgrips or handles shall be permissible for aid in making easy duct connection.

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3.17.2.2 Material and insulation.- The ducts shall be fabricated of a nonpermeable, fuel- and oil-resistant material that will be flexible at all operating ambient temperatures and will withstand all conditions of operation and storage applicable to the air conditioner. Insulation shall be adequately anchored into place.

3.17.2.3 Durability.- The ducts shall be of a type that will withstand rough handling such as encountered when dragged over rough concrete surfaces. The couplings shall withstand repeated droppings onto a concrete surface from a height of 10 feet.

3.17.2.4 Stiffener core.- The ducts shall be of the type that are reinforced by a helical wire stiffener core that tends to extend the duct when released. The wire core shall be covered with a scuff strip that will withstand a 7,000-foot drag over concrete surfaces normally found on runways and taxi strips.

3.17.2.5 Storage provisions.- The ducts shall be compressible to 25 percent of their extended length for storage. Readily accessible container racks shall be provided for duct storage and ease of handling.

3.18 Air filters.- All air intake openings in the air conditioner shall be protected by means of screens of 1/2- to 3/4-inch mesh to prevent the entrance of large foreign objects. Permanent-type washable filters shall be provided to further screen dust and dirt particles from the intake-air openings. The filters shall be coated with a viscous substance. A minimum clearance of 2 inches shall be provided between the screen and the filters.

3.18.1 Cooling-air intake openings.- Filters used in air intake openings supplying only cooling air to components, condensers, et cetera, shall be a 1-inch nominal thickness and shall have a total filter area to permit a maximum face velocity of 750 fpm.

3.18.2 Conditioned-air intake openings.- Filters used in the conditioned-air intake openings shall be of 2-inch nominal thickness and of a total area to permit a maximum face velocity of 500 fpm.

3.18.3 Size.- All filters of the same thickness shall be of the same size.

3.19 Fuel system

3.19.1 Fuel tank.- The air conditioner shall incorporate an integral fuel tank having a 58-gallon capacity. The fuel tank shall incorporate a sump for collection of foreign materials that may settle out of the fuel. The fuel tank shall be vented to the outside of the cabinet.

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3.19.1.1 Sump drain.- A drain, designed to drain overboard, shall be provided in the sump to permit rapid and complete drainage of the fuel tank without tapping the air conditioner.

3.19.1.2 Fill opening.- The fuel tank shall have an external fill opening of not less than 3 inches inside diameter. The fill opening shall be located in an external wall or on top of the cabinet in a location convenient for servicing. The fill opening shall be so recessed below the cabinet surface that the cap will not protrude and shall be so arranged that overflow or spilled fuel will drain overboard and not enter the cabinet. The cap shall be painted red and shall be secured to the tank by means of a safety chain.

3.19.2 Fuel filter.- A 10-micron filter shall be incorporated in the fuel system to remove foreign particles and suspended dirt particles from the fuel passing through the filter. The filter shall be accessible for quick removal for cleaning and replacement purposes.

3.19.3 Fuel gage.- A fuel gage shall be provided to indicate the quantity of fuel in the fuel tank. If an electrical-type gage is used, it shall be so wired that it will not be necessary to start the engine to determine fuel quantity.

3.19.4 Fuel lines.- All fuel lines shall be easily disconnected and removed for servicing. A fuel shutoff valve shall be used to prevent continuous spillage of fuel while disconnecting fuel lines.

3.19.4.1 All fuel lines in the air conditioner shall be constructed of steel or other fire-resistant material. Where flexibility is required in the lines, flexible fire-resistant hose assemblies with couplings shall be used. Hose-clamp connections shall not be used. Connections in metal fuel lines shall not employ soft solder nor any other relatively low-melting-point materials that will be affected by a temperature of 1,250° F.

3.19.5 All gaskets, synthetic rubber seals, et cetera, shall be suitable for use with the specified fuels and shall be satisfactory for use at all ambient and operating temperatures within the scope of the air conditioner.

3.20 Trailer mobility provisions.- The air conditioner shall be mounted on a trailer or integral trailer chassis having mobility provisions in accordance with MIL-M-8090, type II, group C, with the exceptions and additions specified herein.

3.20.1 Frame.- The trailer frame shall be of metal construction and of sufficient size and strength to accommodate all components of the air conditioner.

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3.20.2 Tread width.- The tread width shall be 67 ± 1 inch. The wheel-base-to-tread-width ratio shall be not less than 1.15 to 1.

3.20.3 Springs.- Springs shall be furnished on the trailer to provide good riding qualities, thereby precluding damage to the mounted equipment.

3.20.4 Road clearance.- The fully loaded trailer shall have a road clearance of at least 5-3/4 inches at the differential housing and at least 8 inches at all other locations.

3.20.5 Self-mobility system.- The trailer shall be provided with self-mobility features capable of moving the air conditioner over paved surfaces at low speeds. Tractive power shall be supplied to the two nonsteerable rear wheels.

3.20.5.1 Motor.- The mobility system shall be powered by an electric motor operating from the air conditioner electrical system. The motor shall be of the reversing type suitable for use in propelling the trailer forward or rearward as required. The motor shall be provided with overload protection to prevent high-current draw in event of operation of the self-mobility system off paved surfaces.

3.20.5.2 Batteries.- Two Type 6TN or two Type 6TNC batteries conforming to MS35000-4 or MS24531-1 shall be provided as specified by the procuring activity (see 6.2). An AN2552-3A external power receptacle shall be wired into the normal battery circuit with a reverse-current relay to permit external starting of the engine from an auxiliary power unit.

3.20.5.2.1 Battery compartment.- The battery compartment shall have adequate vents and drains and shall be accessibly located to permit adding water to the batteries without battery removal.

3.20.5.3 The mobility system shall be capable of moving the air conditioner at speeds from 50 to 75 fpm. It shall be possible to move the air conditioner over distances up to 200 feet without starting the prime mover and without detrimental effects to the battery which would prevent satisfactory prime mover starts at temperatures down to 0° F. Provisions shall be made so that the air conditioner can be self-propelled around airfields at the rate specified with the prime mover running.

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3.20.5.4 Declutching mechanism.- An automatic declutching mechanism shall be provided to disconnect the drive train from the wheels when the trailer is being towed. Input torque capacity of the clutch shall be not less than that of the propulsion motor plus that of any intervening gear ratios.

3.20.5.5 Differentials.- Automotive-type differentials shall be provided to properly distribute axle torque.

3.20.5.6 Safe-mobility drive mounting.- The mobility system shall be so designed that flexing of the trailer chassis and frame will not exert excessive strain or twist on the connecting drive components. Provisions shall be made so that the differential housing will have free movement without causing strain or stress on the mobility mounting system that would result in failure.

3.20.5.7 Hubs.- Hubs shall be of a type that will permit the wheels to be removed without removing the hubs.

3.20.5.8 Switches.- The mobility system shall be provided with three pushbutton switches on the towbar adjacent to the lunette eye. Two of the switches shall be located along one side of the towbar, one for forward motion and the other for reverse motion. The third switch shall serve as a master switch and shall require actuation for both forward and reverse operation of the mobility system. The master switch shall be located on the side opposite the other two switches in such a position that it and either the forward or reverse switch can be operated simultaneously. Each switch shall be waterproof and spring-loaded open.

3.21 Control system and instrumentation

3.21.1 Operating controls.- The air conditioner shall incorporate adequate controls for starting and controlling the power of the prime mover and for controlling the cooling capacity, airflow rates, and the air delivery temperatures specified in 3.7.2.5.

3.21.1.1 Prime mover control.- A mechanical engine power control system shall be incorporated using a single lever at the instrument panel. The lever shall have a stop position, a start-idle position, and a run range. Moving the lever from the stop to start-idle position shall first remove the ground from the magneto circuit and then release the idle cutoff. In the start-idle position, the linkage shall control the engine rpm for the best idle speed as determined by the torsional vibration analysis of the engine-component combination. When the throttle lever is in run position, the engine shall be governed at 2,550 rpm. In stopping the engine, moving the lever from run to start-idle shall return the engine to the best idle speed. Moving from start-idle to stop shall first actuate the idle cutoff and then ground the magneto circuit. It shall not be possible to engage the engine starter when the lever is in either the stop or run position or when the engine is running.

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3.21.1.2 Airflow controls.- A dump valve shall be provided to manually control airflow so that any rate of airflow between 20 and 100 percent of that specified in 3.7.2.1, at maximum delivery pressure or free delivery, can be obtained at governed rpm (2,550) of the engine. In addition, the air conditioner shall have an automatic airflow control that is sensitive to discharge air temperature to regulate the discharge temperature under the following conditions:

a. At extremely high humidity conditions; for example, in 3.7.2.3 where a reduced flow rate is required unless the air conditioner is capable of the additional capacity required at maximum flow rate and 45° F delivery temperature

b. At low (or free) delivery pressures if the flow rate with open manual controls increases to the extent that delivery temperature exceeds 45° F.

3.21.1.3 Controls shall be provided to manually vary the air delivery temperature from the normal range (35° to 45°) up to 65° F at any airflow rate and discharge pressure.

3.21.1.4 Refrigerant system controls.- The refrigerant system shall be equipped with controls to automatically regulate the air conditioner capacity in accordance with the requirements specified herein.

3.21.1.4.1 Pump-down control.- A control shall be provided for automatic pump down of the refrigerant system. This shall be accomplished by use of a solenoid shutoff valve located in the liquid refrigerant line between the receiver and expansion valve. Subsequent to shut off of this valve, the compressor shall operate until the compressor suction pressure is reduced to a value that will actuate the low-pressure cutoff switch. It shall be possible to stop the prime mover and to operate it in the start-idle and run ranges without these operations being affected by the automatic pump-down control.

3.21.2 Safety controls.- Adequate safety controls shall be provided to accomplish at least the following functions:

a. Prevent prime-mover starting when inlet or outlet doors are in the closed position. (When the air conditioner is in operation, these doors must be in the open position to prevent damage to the air conditioner.)

b. Ground the prime mover magneto in case of low oil pressure, prime-mover overheating, or excessive temperatures or pressures of critical components.

c. Ground the magneto when the refrigerant suction pressure is reduced to a value between 20 and 15 psi.

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3.21.3 Instrumentation

3.21.3.1 Instrument and control panel.- The air conditioner shall be provided with a composite instrument and control panel on which all instruments, thermometers, throttle levers, switches, warning lights, et cetera, shall be mounted (within the limits of practicability). The instrument and control panel and accessories shall be readily accessible to operating personnel and all items on the panel shall be permanently and legibly marked. The panel shall be covered by a weathertight door and, if necessary, the panel and instruments shall be shockmounted. The panel shall be illuminated by a switch-actuated white light. The bottom of the instrument panel shall be at least 48 inches from ground level.

3.21.3.2 Instruments.- Instruments indicating the following shall be among those included on the instrument panel:

a. Outlet air temperature (°F)

b. Discharge pressure (psi)

c. Approximate airflow rate (pm)

d. Prime-mover speed

e. Prime-mover manifold pressure

f. Prime-mover cylinder head temperature

g. Prime-mover operating time

h. Prime-mover oil pressure

i. Head pressure (refrigerant system)

j. Suction pressure (refrigerant system).

3.21.3.3 Hourmeter.- The hourmeter shall have a range of 0 to 9,999 hours and shall be a cumulative-indicating, dial-face type that cannot be reset to 0 hours without a special tool or special servicing. The hourmeter may be electrically powered and actuated by ignition or oil pressure, or it may be tachometer driven and geared in such a manner that 1 hour is registered in 1 hour of running at normal operating speed of the prime mover at design conditions (see 3.7.2.1).

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3.22 Accessory equipment

3.22.1 Exhaust stack.- A removable extension exhaust stack shall be provided and shall extend 18 inches above the top of the cabinet. The exhaust stack shall be stored within the cabinet when not in use. The cabinet shall be provided with a cover for protection of the internal exhaust pipe when the extension is removed. As an alternate, an extensible exhaust stack 18 inches in length may be used. The stack shall be designed to be readily extended and lowered and shall include a handle, rollers, and latch for the extended position, and a cover for use when the stack is in the lowered position.

3.22.1.1 Muffler system.- A muffler system shall be provided to reduce prime-mover exhaust noise to a value within the noise level limitations specified in 3.6.7 but not create an excessive backpressure on the prime mover that will appreciably effect performance. To minimize the danger of discharging large sparks from the exhaust stack, the muffler shall be of a type that tends to eliminate sparks.

3.22.2 Handbook and record compartment.- A compartment shall be provided as an integral part of the air conditioner to contain equipment operation books and maintenance and inspection records. The compartment shall be a minimum size of 9 by 11-1/2 by 2 inches and shall be weatherproof if not contained within the cabinet.

3.22.3 Water separation.- Provisions shall be incorporated so that at least 90 percent of the water resulting from cooling the conditioned air is removed from the air, collected, and drained overboard from the air conditioner through a drain line when the unit is operated at an angle of tilt up to 8-1/2° from the true horizontal in any direction.

3.23 Part numbering of interchangeable parts.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern manufacturer's part numbers and changes thereto.

3.24 Dimensions.- The dimensions of the air conditioner (exclusive of towbar, pintle hook, and removable tiedown rings) shall not exceed the following:

Dimension	Inches
Width	76
Height	72
Length	104

3.24.1 Dimensions and tolerances.- Dimensions and tolerances not specified shall be consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the air conditioner, they shall be held or limited accordingly.

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3.25 Weight.- The weight of the air conditioner without fuel shall not exceed 5,800 pounds.

3.26 Color.- The color of the air conditioner shall be chrome yellow conforming to FED-STD-595, color 13538. This shall apply to all parts except lubrication fittings, tiedown rings, and other plated parts that are externally visible. Internal components need not conform to this color.

3.27 Finishes and protective coatings

3.27.1 Magnesium-alloy parts.- Magnesium base alloy parts shall be pretreated in accordance with MIL-M-3171, type IV anodic coating, followed by two coats of zinc-chromate primer conforming to MIL-P-8585. Surfaces not exposed to weather shall be coated with the primer and one coat of enamel conforming to TT-E-489. Surfaces directly exposed to the elements or to wear by handling shall be given a second coat of the aforementioned enamel.

3.27.2 Aluminum-alloy parts

3.27.2.1 Protective coatings.- Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.27.2.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.27.2.1.2 Where the primary purpose of the treatment is to afford a suitable paint base, chemical films conforming to MIL-C-5541 may be used in lieu of anodizing.

3.27.2.1.3 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film conforming to MIL-C-5541 in lieu of anodizing.

3.27.2.1.4 Where abrasion resistance is a factor, chemical films conforming to MIL-C-5541 shall not be used in lieu of anodizing.

3.27.2.2 Inside surfaces.- After the treatment specified in 3.27.2.1, aluminum-alloy surfaces not exposed to wear or elements shall be painted with one coat of primer conforming to MIL-P-8585 and one coat of enamel conforming to TT-E-489.

3.27.2.2.1 Exposed surfaces.- Aluminum-alloy surfaces directly exposed to the elements or to wear by handling shall be painted with the primer and two coats of the enamel specified in 3.27.2.2.

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3.27.3 Ferrous parts.- Bolts or parts that are cadmium plated shall be processed in accordance with QQ-P-416, of a type and class that are adequate to achieve the degree of protection required.

3.27.3.1 Surfaces of ferrous parts to be painted shall be sandblasted or pickled to produce a clean surface and painted with one coat of primer conforming to TT-P-664. Surfaces directly exposed to wear or the elements shall be painted with two coats of enamel conforming to TT-E-489. Internal surfaces shall be covered with one coat of this enamel.

3.27.4 On surfaces where enamel is specified in accordance with TT-E-489, two coats of lacquer conforming to MIL-L-7178 may be substituted. Where one coat of enamel is specified, two coats of lacquer shall be used. However, if lacquer is used, the primer specified in MIL-P-8585 shall be substituted for the primer specified in TT-P-664.

3.28 Marking

3.28.1 Starting and operating instructions.- Brief instructions for starting and operating the air conditioner together with a wiring diagram of the air conditioner shall be located on the internal surface of the instrument-panel door. The instructions and diagram shall be such that oil, dirt, water, et cetera, will not affect their legibility nor cause them to be obliterated. The wiring diagram shall conform to MIL-D-70327.

3.28.2 Tire pressure.- The following information shall be painted on the forward half of each side of the trailer frame in 1-inch-high, black, block-type letters:

TIRE PRESS. 30 LBS.

3.28.3 Spreader bars.- The following information shall be located near the hoisting and tiedown provisions in 1-inch-high, block-type letters:

SPREADER BARS SHOULD BE USED WHEN HOISTING

3.28.4 Pintle marking.- The following information shall be stenciled above the pintle:

WARNING - DO NOT TOW MORE THAN ONE SIMILAR VEHICLE FROM PINTLE BELOW. DO NOT TOW ANY VEHICLES WHEN OPERATING ON SELF-POWER.

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3.28.5 Towbar markings

3.28.5.1 Mobility system switches.- The three pushbutton switches specified in 3.20.5.8 shall be marked as follows:

3.28.5.1.1 The two switches provided along one side of the towbar shall be marked as follows, as applicable:

FORWARD
REVERSE

3.28.5.1.2 The third or master switch shall be marked as follows:

SELF MOBILITY

3.28.5.2 Towbar.- A warning plate or stencil containing the following information in bold letters shall be placed adjacent to the three pushbutton switches on the towbar adjacent to the lunette eye:

DO NOT USE SELF-MOBILITY OVER 200 FEET WITHOUT STARTING ENGINE.

3.29 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The air conditioner shall be identified with a securely attached nameplate marked with the information specified in MIL-S-8512.

3.30 Marking for identification.- Unless otherwise specified (see 6.2), metals and metal products shall be marked in accordance with the following standards, as applicable:

Products	Applicable Standard
a. Selected nickel and nickel alloy	FED-STD-182
b. Selected iron and steel	FED-STD-183
c. All aluminum, magnesium, and titanium	FED-STD-184
d. All copper and copper base alloy	MIL-STD-185

3.31 Government-furnished property.- When specified in the contract (see 6.2), the Government will furnish the following to the contractor, upon his request, for use in making the air conditioner:

Stock No.	Item Description	Quantity
2805-633-6689	Packette engine GLU-3/U conforming to MIL-E-26248 and MS24559	1

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3.31.1 Government-loaned property.- When specified in the contract (see 6.2), the Government will loan the following to the contractor upon his request:

Stock No.	Item Description	Quantity
4120-141-9130	Type MA-3 air conditioner	1
4520-219-7969 or 4520-219-7968	Type H-1 heater conforming to MIL-H-4607	1

3.32 Workmanship.- The air conditioner, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.32.1 Fabrication.- Boxes, cases, shields, and compartment walls shall be fabricated by casting, drawing or bending, and welding except that riveting or bolting may be used where ease of servicing of the air conditioner requires that a removable panel construction be used or where the applicable stresses dictate the use of a strong aluminum alloy that does not provide a good weld.

3.32.2 Gears.- Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, and other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.32.3 Riveting.- Riveting operations shall be performed to insure tightness and satisfactory heading of rivets.

3.32.4 Cleaning.- The air conditioner shall be thoroughly cleaned and loose, spattered, or excess solder; metal chips; and other foreign material removed during and after final assembly.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of the air conditioner shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5

4.3 Test conditions

4.3.1 Apparatus.- Insofar as practicable, apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to insure laboratory accuracy.

4.3.2 Revolutions per minute.- Whenever practicable, rotational speed shall be determined by means of a directly coupled positive counter that will actually count the revolutions for a period of not less than 1 minute. A combination unit that counts the revolutions while measuring the time required in connection with other data is preferred. Strobotachometers may be used provided calibration data can be furnished to attest to their accuracy. The use of hand tachometers will not be acceptable.

4.3.3 Airflow.- Airflow shall be measured by means of smooth-approach nozzles or flat-plate orifices conforming to ASME Code 19.5:4. Other apparatus, such as pitot tubes, rotameters, et cetera, may be used only when specifically approved by the procuring activity. The rate of airflow shall be expressed in terms of pounds per minute.

4.3.4 Fuel consumption.- Fuel consumption measurement shall be made by either the volume or weight method. The quantity selected for the volume or weight method shall be such that each reading will cover an elapsed time of at least 1 minute. Flowmeter readings for fuel consumption shall be used only when the flowmeter and use thereof have been specifically approved by the procuring activity. Regardless of the method used, all expression of quantities shall be recorded on a weight basis.

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4.3.5 Humidity.- Humidity shall be determined by means of a psychrometric chart from readings of wet- and dry-bulb temperatures. Air velocity over the wet-bulb thermometer shall be sufficient to insure reasonable accuracy. These readings shall be taken hourly. The thermometers shall be so arranged that the air passing over the wet bulb will not subsequently pass over the dry bulb. The wick of the wet bulb shall be kept wet with distilled water only and shall be replaced whenever a perceptible quantity of oil, dirt, incrustation, or other foreign substance has accumulated thereon.

4.3.6 Barometric pressure.- Barometric pressure shall be measured twice in 24 hours by means of a mercurial barometer at the same elevation as the test. The pressure shall be corrected for the temperature of the scale and of the mercury. Readings obtained from aneroid barometers will not be acceptable.

4.3.7 Temperatures.- Temperatures shall be measured by means of appropriately located thermocouples and properly calibrated potentiometers. Insofar as practicable, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit. Laboratory-type mercury thermometers shall be permissible if used correctly in the application.

4.3.8 Pressure.- Insofar as practicable, pressure and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. The gages shall be of laboratory type having a dial diameter of not less than 4 inches.

4.3.9 Oil consumption.- Oil consumption shall be determined by accurately measuring the quantity necessary to maintain a given level in the oil reservoir.

4.3.10 Preliminary run-in.- Any preliminary run-in specified by the manufacturer shall be performed prior to submission of the air conditioner for testing.

4.4 Preproduction testing

4.4.1 Preproduction test samples tested by the contractor.- The contractor shall subject two air conditioners to the preproduction tests specified in 4.4.3.

4.4.2 Preproduction test report.- After the contractor completes the preproduction tests, he shall prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

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4.4.3 Preproduction tests.- The preproduction test samples shall be subjected to the following tests as described under 4.6:

Sample No. 1	Sample No. 2
Examination of product	Examination of product
Operational tests	Operational tests
Mobility tests	Tilted position
Performance and endurance	Acceleration forces
Radio interference	Duct tests
Noise level test	Environmental tests

4.5 Acceptance tests.- Acceptance tests shall consist of:

a. Individual tests

b. Periodic sampling plan and tests.

4.5.1 Individual tests.- Each air conditioner shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Operational tests . . . . . See 4.6.2.

4.5.2 Periodic sampling plan and tests.- One air conditioner shall be selected at random from the first 10, one from the next 40, and one from each subsequent 50 or fraction thereof produced and subjected to the test specified in 4.6.3.

4.6 Test methods

4.6.1 Examination of product.- The air conditioner shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.6.2 Operational tests.- The air conditioner shall be subjected to and shall satisfactorily complete the following tests:

4.6.2.1 Air-conditioning unit.- After the temperature of the air-conditioning unit has been stabilized, the air-conditioning unit shall be operated for 30 minutes at the prevailing normal room or ambient temperature. The measured or indicated airflow shall be not less than 100 pm and the discharge temperature shall not exceed 50° F. The temperature, airflow, and safety controls shall be checked for operational stability.

4.6.2.2 Trailer.- A functional test on all operating components, such as brakes, steering, lights, et cetera, shall be conducted to insure proper operation.

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4.6.2.3 Three-mile road test.- The complete air conditioner shall be towed for a distance of 3 miles to determine general functioning of all the chassis components. The unit shall be inspected for excessive vibration, tracking, distortion, loosening, and maladjustment during this test. After testing, no signs of malfunctioning shall be evident.

4.6.3 Acceptance performance test.- Under the ambient conditions specified in 3.7.2.1, it shall be demonstrated that within 10 minutes after prime mover start the air conditioner is capable of delivering 100 pm of air at less than 50° F and at static pressures ranging from 0 to 3 psig. The air conditioner shall be operated for 1 hour under these ambient conditions and shall continue to meet the requirements specified in 3.7.2.1. At the end of this test, all safety controls shall be checked for effectiveness. The air conditioner shall exhibit no malfunction, leaks, rough operation, vibration, or other irregularities as a result of this test.

4.6.4 Mobility tests.- Prior to the tests specified in 4.6.5, the air conditioner shall be subjected to mobility tests. The air conditioner shall be towed by a truck as follows:

a. Over a highway at a speed of not less than 20 mph for a distance of not less than 100 miles

b. Over cross-country terrain at a speed of not less than 10 mph for a distance of not less than 70 miles

c. Over a course consisting of obstructions at least 1 inch in height regularly spaced at 6-foot intervals and at right angles to the direction of travel of the air conditioner at a speed of 10 mph for 1/2 hour. The tire pressure for this test shall be 30 psi minimum.

4.6.4.1 Maneuverability.- The air conditioner shall be towed through 10 complete turns each to the right and left at a speed of 8 to 10 mph over a dry concrete or macadam surface. The trailer shall turn within a 17-1/2-foot radius without capsizing or skidding.

4.6.4.2 Self-propulsion test.- The air conditioner, equipped with a fully charged battery, shall be self-propelled over a level, paved surface for 200 feet at prevailing ambient temperature. This shall be followed by 10 prime mover starts within 10 minutes to demonstrate the remaining effectiveness of the battery. An analysis shall be made as to the ability of the self-mobility system and battery to meet the requirements of 3.20.5.3 at 0° F.

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4.6.4.2.1 The air conditioner shall be self-propelled over a paved surface, with the prime mover running, for a distance of 10 miles. At the end of each mile, the air-conditioner prime mover shall be stopped for 15 minutes and restarted. After the 10 miles have been traveled in this manner, the prime mover shall be started 10 times in 10 minutes to demonstrate that no detrimental effects have occurred.

4.6.5 Performance and endurance tests (500 hours).- The air conditioner shall pass a performance test and an endurance test of 500 hours total duration. The test time shall be divided as follows:

a. Performance test of 100 hours

b. Endurance test of 400 hours.

4.6.5.1 Performance test (100 hours).- The conditions for this phase of the test shall be as follows:

a. Basic rating - 100° F DB, 85° F WB. The air conditioner shall be operated for 20 hours. Performance shall be recorded at various air delivery pressures between 0 and 3 psig at full capacity as specified in 3.7.2 and at various reduced flow rates.

b. Design load - 100° F DB, 76° F WB. The air conditioner shall be operated for 20 hours. Performance data shall be recorded at various air delivery pressures between 0 and 3 psig at full capacity as specified in 3.7.2.1 and at various reduced flow rates.

c. Hot-day ambient conditions - 125° F DB, 75° F WB. The air conditioner shall be operated for 10 hours. Performance data shall be recorded at various air delivery pressures between 0 and 3 psig at full capacity as specified in 3.7.2.2 with various reduced flow rates.

d. Humid-day conditions - 97° F DB, 87° F WB. The air conditioner shall be operated for 10 hours. Performance data shall be recorded at various air delivery pressures between 0 and 3 psig at full capacity as specified in 3.7.2.3 and at various reduced flow rates. During this test, it shall also be demonstrated that the air conditioner meets the requirements of 3.7.2.5 and 3.22.3.

e. Various operating conditions - Ambient operating range shall be as specified in 3.7.1 except that temperatures exceeding 125° F DB or temperatures less than 60° F DB shall not be used. The air conditioner shall be operated at various ambient temperatures and relative-humidity conditions within this range for 40 hours. Data shall be recorded during part of the test at maximum delivery pressure of 3 psig and during the remainder of the test shall be recorded at approximately free air delivery. Data shall be recorded at airflow delivery rates varying from maximum to minimum under these varying ambient conditions. During this test, it shall be demonstrated that the air conditioner meets the requirements of 3.7.2.4. During this test, it shall also be demonstrated that the air conditioner meets the pressure relief requirements of 3.13.

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4.6.5.1.1 The cycling of the tests specified in 4.6.5.1 shall be at the option of the testing activity, but the total continuous hours shall amount to those specified. One cycle shall be not less than a duration of 1 hour.

4.6.5.1.2 The test specified in 4.6.5.1e shall be conducted to provide sufficient data to arrive at complete performance characteristics of the air conditioner for the specified ambient ranges of 3.7.1a, b, and c.

4.6.5.2 Endurance test (400 hours).- The air conditioner shall be operated for an additional 400 hours at design-load prime-mover speed, full air delivery of the air conditioner, and at delivery pressures as follows: 100 hours at 0, 1, 2, and 3 psig. Prevailing ambient conditions shall be used provided the dry-bulb temperature exceeds 75° F, wet-bulb temperature exceeds 65° F, and relative humidity exceeds 30 percent. A performance check shall be conducted on the air conditioner under the conditions specified in 3.7.2.1 at intervals of 100 hours of operation and at the completion of the 500-hour testing period. Upon completion of the 500-hour performance-endurance tests, the air conditioner shall fulfill the performance requirements specified in 3.7.2.1 The performance check shall be of sufficient duration to record all necessary data to ascertain satisfactory operation, but shall not be in excess of 5 hours.

4.6.6 Radio interference.- The complete air conditioner shall be tested for radiated radio-interference suppression in accordance with MIL-I-26600 but limited to frequencies from 0.15 mc up to and including 150 mc for radiated interference. When a shielded room is not readily available, a test location shall be selected in lieu thereof. The inherent background level of the test location shall be within the limits specified in MIL-I-26600. The meter and test specimen shall have a common ground potential in substitution for the ground plane. The antenna location shall be limited to 6 feet from the perimeter of the test specimen.

4.6.7 Noise level.- The criteria specified in 3.6.7 gives the levels for the operator and for the average at 25 feet. The levels for the operator are the average levels measured at head level (5 ft 4 in.) in the area where the operator is located. This measurement shall be made by slowly moving the measuring microphone in a circle 2-feet in diameter centered on the most probable position for an operator. The average level for each octave band shall be determined by a simple arithmetic average of the minimum and maximum noise levels (in db) found on this circle. The average levels at 25 feet shall be determined from measurements at head level on a circle 25 feet in radius, centered on the geometrical center of the unit. The noise level for each octave band shall be determined at 12 positions on the circle equally spaced every 30°. The average noise level for each band shall be calculated by a simple arithmetic average of the 12 noise levels (in db) for the band. Measurements shall be made with the unit operating at the

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maximum capacity, consistent with the prevailing ambient, for cooling and for heating modes of operation as a minimum. All acoustic measurements shall be made with equipment in accordance with ASA Bulletins Z24.3 and Z24.10. The flat frequency response (C weighing network) of the sound-level meter shall be used in making the measurements.

4.6.8 Tilted position.- The air conditioner shall be tilted in four different positions, 90° apart, at an angle of 8-1/2° to the true horizontal plane and checked for satisfactory operation of all components during tilt.

4.6.9 Acceleration forces.- The air conditioner shall be attached to a pallet in the manner specified for loading in an aircraft. The palletized air conditioner shall be slid or rolled down an inclined plane or otherwise accelerated and stopped in such a manner as to produce the accelerations specified in 3.6.6 as measured by an accelerometer mounted on the pallet. Evidence of permanent deformation of structural members or failure to operate and meet the requirements of this specification due to the 3g tests shall be cause for rejection. When subjected to the 8g and 4-1/2g tests, the major components shall not break loose but the air conditioner need not be serviceable. Following these tests, the air conditioner shall be examined, operated, and subjected to performance tests specified herein.

4.6.9.1 Hoisting provisions.- The suitability of the hoisting provisions specified in 3.6.5 shall be demonstrated. The air conditioner shall be hoisted from a single point by attaching four equal-length cables, opposite ends terminating at the common point, at the air conditioner. While statically suspended, the air conditioner shall not tilt at an angle of more than 15° with the horizontal. The air conditioner shall also be hoisted to demonstrate compliance with the vertical acceleration requirement of 2g.

4.6.10 Duct tests.- The external ducts shall be subjected to the following tests. No breakage, cracking, tearing, loosening of connections, or other damage that would render the ducts inoperative shall result from these tests.

4.6.10.1 Internal pressure.- One of the ducts specified in 3.17.2 shall be connected to mating male and female couplings that are sealed, except the male coupling shall be provided with a connection for an air supply. The duct shall be subjected to an internal pressure of 6 psig. Measurement of the flow required to maintain 6 psig will provide the rate of leakage. The total air leakage through the duct fabric and couplings shall not exceed 2 pm.

4.6.10.2 Tensile pull.- The ducts shall be subjected to a tensile pull of 150 pounds with the force applied to the end connections.

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